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Chapter 63 - Performance Work On A Real Port

Performance work on a large IE port is measurement work. Guessing is not enough when one frame can involve game code, matrix work, texture streams, Voodoo command submission, audio voice writes, file activity, and input.

63.1 Count The Frame

The case-study port records counters for the work that matters:

Counter family	What it answers
Triangles and texture rectangles	How much picture work reached Voodoo
Draw calls	How often the drawing contract was invoked
MMIO writes	How much register traffic the frame generated
Command submits and pairs	How much work moved through Voodoo streams
Texture stream bytes	How much texture data crossed into Voodoo
Audio voice writes	How much audio control traffic was emitted
Clipped triangles	How much graphics work was rejected before drawing
Graphics-worker frames and time	Whether translation or presentation owns the frame
Audio-worker pumps and time	Whether sequence work keeps its cadence
Coprocessor drains and waits	Whether useful overlap was achieved

These are not decorative numbers. They tell you whether a change moved work out of the hot path or merely moved it somewhere harder to see.

63.2 Compare Like With Like

Frame-rate readings are useful only when the scene, frame group, and instrumentation are known. A short race scene, a title screen, and a blank smoke frame do not measure the same thing.

The case study uses groups of frames rather than single-frame claims. It also separates time spent by the main M68K, graphics worker, IE64 TnL worker, audio worker, and Voodoo submission path. That makes changes such as command streams, texture upload paths, audio shadowing, and coprocessor batching visible.

63.3 Reduce Traffic, Not Meaning

Several optimisations preserve the same visible behaviour:

- Keep texture copies in high RAM, retain uploaded generations in Voodoo slots, and bind unchanged generations by identifier; retransmit them when slots are not available.
- Submit Voodoo register writes through a command stream.
- Avoid rewriting unchanged audio voice fields.
- Submit one coarse frame contract to the graphics worker.

- Batch transform and lighting work for IE64.
- Keep only one graphics frame and one audio pump in flight, so overlap does not become unbounded ownership.
- Measure MMIO volume before and after each change.

The work is still the same game frame. The bus traffic is cleaner.

63.4 The General IE Lesson

Profile the contract, not your hopes. Count triangles, bytes, MMIO writes, voice updates, command pairs, worker completions, and frame groups. Attribute time to the pipeline stage that owns it, then optimise the path that the measurements prove is hot.